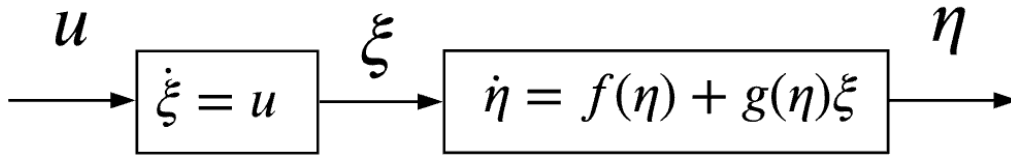


Backstepping control

BASIC CONCEPTS

$$\begin{aligned}\dot{\eta} &= f(\eta) + g(\eta)\xi \\ \dot{\xi} &= u\end{aligned}$$



The problem: design u , s.t. $\boxed{(\eta, \xi) \rightarrow (0,0)}$ as $t \rightarrow \infty$

Step 1: [choose PHI fcn]

Let $\xi = \phi(\eta)$, [st $\boxed{\phi(\eta) = 0 \leftrightarrow \eta = 0}$] [it has only one zero and it is at the origin]

$$\dot{\eta} = f(\eta) + g(\eta)\xi \rightarrow \dot{\eta} = f(\eta) + g(\eta)\phi(\eta)$$

What is the condition on $\phi(\eta)$ for the η dynamics to be stable [A.S.] ?

Let's use a Lyapunov fcn $V_1(\eta)$ for that,

[lyap fcn must be pos-def-fcn], [time-der of lyap fcn must be neg-def-fcn]

$$\frac{d}{dt}V_1(\eta) = \frac{\partial V_1(\eta)}{\partial \eta} \dot{\eta} = \frac{\partial V_1(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)]$$

$$\boxed{\frac{d}{dt}V_1(\eta) < 0} \text{ or } \boxed{\frac{d}{dt}V_1(\eta) < -W(\eta)}, W(\cdot) \text{ is a pos-def-fcn}$$

$$\frac{d}{dt}V_1(\eta) < -W(\eta)$$

$$\boxed{\frac{\partial V_1(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < -W(\eta)}$$

So $V_1(\eta)$ must satisfy these two constraints:

$$\boxed{\begin{aligned}V_1(\eta) &> 0 \\ \frac{\partial V_1(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] &< -W(\eta)\end{aligned}}$$

Step 2: [change of variables]

Define

$$z = \xi - \phi(\eta) \rightarrow \xi = z + \phi(\eta)$$

and

$$\begin{array}{ccccc} \dot{\eta} = f(\eta) + g(\eta)\xi & \rightarrow & \dot{\eta} = f(\eta) + g(\eta)[z + \phi(\eta)] & \rightarrow & \dot{\eta} = [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z \\ \xi = u & & \xi = u & & \xi = u \end{array}$$

What is z-dyns?

$$z = \xi - \phi(\eta) \rightarrow \dot{z} = \dot{\xi} - \frac{d}{dt}\phi(\eta) \rightarrow \dot{z} = \dot{\xi} - \frac{\partial\phi(\eta)}{\partial\eta}\dot{\eta}$$

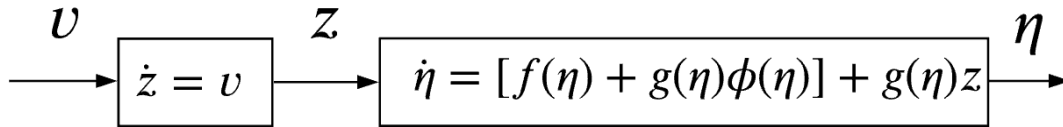
$$\dot{z} = \dot{\xi} - \frac{\partial\phi(\eta)}{\partial\eta}\dot{\eta} \rightarrow \dot{z} = u - \frac{\partial\phi(\eta)}{\partial\eta}\dot{\eta}$$

In the new coordinate system, the dyns,

$$\begin{array}{ccc} \dot{\eta} = [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z & \rightarrow & \dot{\eta} = [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z \\ \dot{z} = u - \frac{\partial\phi(\eta)}{\partial\eta}\dot{\eta} & & \dot{z} = u - \dot{\phi} \end{array}$$

Define a new input signal $\boxed{v = u - \dot{\phi}}$, therefore the new dyns,

$$\begin{array}{ccc} \dot{\eta} = [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z & \rightarrow & \dot{\eta} = [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z \\ \dot{z} = u - \dot{\phi} & & \dot{z} = v \end{array}$$



Now the new problem is as the one given in the block-diagram. the second subsystem is A.S., when zero "input" [z-signal] is applied.

All of the transformations and new variable definitions were introduced to make the second subsystem A.S. And This has consequences, now we can find u using lyap-fcn in this coordinate frame. [the reason why we were not able to find u in the original coordinates by using lyap fcn approach is that if we did we would have $\frac{fcn(\eta)}{\xi}$ terms in our ctrl law and as xi goes to zero, these terms becomes problematic]

Continuing, the new problem, design v st $\boxed{(\eta, z) \rightarrow (0,0)}$

$$\begin{array}{c} \dot{\eta} = [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z \\ \dot{z} = v \end{array}$$

Choose a lyap-fcn as

$$V_2(\eta, z) = V_2(\eta, [\xi - \phi(\eta)]) = V_2(\eta, \xi) = V_{1x}(\eta) + \frac{1}{2}z^2$$

$V_2(\eta, \xi)$ must be pos-def-fcn, and $V_{1x}(\eta)$ must be pos-def-fcn. $\boxed{\frac{d}{dt}V_2(\eta, z)}$ must be neg-def-fcn.

Take the time-der

$$\frac{d}{dt}V_2(\eta, z) = \frac{d}{dt}\left[V_{1x}(\eta) + \frac{1}{2}z^2\right] = \frac{\partial V_{1x}(\eta)}{\partial \eta}\dot{\eta} + z\dot{z} < 0$$

$$\frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] + g(\eta)z + z\dot{z} < 0$$

$$\frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] + \frac{\partial V_{1x}(\eta)}{\partial \eta} [g(\eta)z] + z\dot{z} < 0$$

$$\frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] + \frac{\partial V_{1x}(\eta)}{\partial \eta} [g(\eta)z] + zv < 0 \rightarrow \boxed{\begin{array}{l} \frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < 0 \\ \frac{\partial V_{1x}(\eta)}{\partial \eta} [g(\eta)z] + zv < 0 \end{array}}$$

So, $V_{1x}(\eta)$ must satisfy these two constraints: $[W_x(\eta)]$ is a pos-def-fcn

$$\boxed{\begin{array}{l} V_{1x}(\eta) > 0 \\ \frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < -W_x(\eta) \end{array}} \rightarrow \boxed{\begin{array}{l} V_{1x}(\eta) > 0 \\ \frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < 0 \end{array}}$$

Let us recall the $V_1(\eta)$ fcn that we have constructed to determine $\phi(\eta)$ fcn,

$$\boxed{\begin{array}{l} V_1(\eta) > 0 \\ \frac{\partial V_1(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < -W(\eta) \end{array}}$$

Therefore, V_{1x} fcn can be selected as

$$V_{1x}(\eta) = V_1(\eta)$$

Let us go back to lyap-fcn-constraints to design v ,

$$\boxed{\begin{array}{l} \frac{\partial V_{1x}(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < 0 \\ \frac{\partial V_{1x}(\eta)}{\partial \eta} [g(\eta)z] + zv < 0 \end{array}} \rightarrow \boxed{\begin{array}{l} \frac{\partial V_1(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] < -W(\eta) \\ \frac{\partial V_1(\eta)}{\partial \eta} [g(\eta)z] + zv < 0 \end{array}}$$

So the cons-2 can be changed by another cons,

$$\boxed{\frac{\partial V_1(\eta)}{\partial \eta} [g(\eta)z] + zv = -kz^2} \rightarrow \boxed{\frac{\partial V_1(\eta)}{\partial \eta} [g(\eta)z] + zv < 0}$$

Solving for v yields,

$$\boxed{\frac{\partial V_1(\eta)}{\partial \eta} [g(\eta)z] + zv = -kz^2} \rightarrow \boxed{\frac{\partial V_1(\eta)}{\partial \eta} [g(\eta)] + v = -kz} \rightarrow \boxed{v = -kz - \frac{\partial V_1(\eta)}{\partial \eta} [g(\eta)]}$$

Therefore,

$$\frac{d}{dt} V_2(\eta, z) \leq -W(\eta) - kz^2$$

Which indicates the stability of the system.

$$v = u - \dot{\phi} \rightarrow u = v + \dot{\phi} \rightarrow u = v + \frac{\partial \phi(\eta)}{\partial \eta} \dot{\eta} \rightarrow u = v + \frac{\partial \phi(\eta)}{\partial \eta} [f(\eta) + g(\eta)\xi]$$

$$u = \left[-\frac{\partial V_1(\eta)}{\partial \eta} g(\eta) - kz \right] + \frac{\partial \phi(\eta)}{\partial \eta} [f(\eta) + g(\eta)\xi]$$

$$\boxed{u = \left[-\frac{\partial V_1(\eta)}{\partial \eta} g(\eta) - k[\xi - \phi(\eta)] \right] + \frac{\partial \phi(\eta)}{\partial \eta} [f(\eta) + g(\eta)\xi]}$$

CTRL ALGO: [backstepping]

$$\boxed{\begin{array}{l} \text{Find } \phi(\eta) \text{ st} \\ \exists V_1(\eta) > 0, \exists W(\eta) > 0 \text{ st} \\ \frac{\partial V_1(\eta)}{\partial \eta} [f(\eta) + g(\eta)\phi(\eta)] \leq -W(\eta) \end{array}}$$

This is the general problem definition, but easier version can be obtained by assuming,

$$V_1(\eta) = \frac{1}{2}\eta^2 \rightarrow \eta[f(\eta) + g(\eta)\phi(\eta)] \leq -W(\eta)$$

Again, by assuming,

$$W(\eta) = \eta^2$$

$$\eta[f(\eta) + g(\eta)\phi(\eta)] = -\eta^2 \rightarrow \text{solve for } \phi(\eta)$$

PROBLEM 1

Hassan Khalil – nonlinear systems 3rd Ed Example 14.9

$$\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = u \end{array} \quad \text{Find } u(x_1, x_2, x_3) \text{ s. t. } \boxed{\dot{x} = f_{CL}(x)} \text{ is A.S.}$$

$\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = u \end{array}$	$\begin{array}{l} x_2 = \phi_1(x_1) + z_2 \\ \dot{x}_2 = \dot{\phi}_1 + \dot{z}_2 \\ \dot{z}_2 = \dot{x}_2 - \dot{\phi}_1 \\ \text{Subs } \boxed{\dot{x}_2 = x_3} \\ \dot{z}_2 = x_3 - \dot{\phi}_1 \end{array}$	$\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + [\phi_1 + z_2] \\ \dot{z}_2 = x_3 - \dot{\phi}_1 \\ \dot{x}_3 = u \end{array}$
$\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + [\phi_1 + z_2] \\ \dot{z}_2 = x_3 - \dot{\phi}_1 \\ \dot{x}_3 = u \end{array}$	$\begin{array}{l} x_3 = \phi_2(x_1, x_2) + z_3 \\ \dot{x}_3 = \dot{\phi}_2 + \dot{z}_3 \\ \dot{z}_3 = \dot{x}_3 - \dot{\phi}_2 \\ \text{Subs } \boxed{\dot{x}_3 = u} \\ \dot{z}_3 = u - \dot{\phi}_2 \end{array}$	$\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + [\phi_1 + z_2] \\ \dot{z}_2 = [\phi_2 + z_3] - \dot{\phi}_1 \\ \dot{z}_3 = u - \dot{\phi}_2 \end{array}$

$$\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + [\phi_1 + z_2] \\ \dot{z}_2 = [\phi_2 + z_3] - \dot{\phi}_1 \\ \dot{z}_3 = u - \dot{\phi}_2 \end{array} \rightarrow V(x_1, z_1, z_2) = V_1(x_1) + V_2(z_2) + V_3(z_3)$$

By choosing, $V_1(x_1) = \frac{1}{2}(x_1)^2$, $V_2(z_2) = \frac{1}{2}(z_2)^2$, $V_3(z_3) = \frac{1}{2}(z_3)^2$

[NOTE: more sophisticated functions can also be chosen]

$$\frac{d}{dt}V(x_1, z_1, z_2) = \frac{d}{dt}[V_1(x_1) + V_2(z_2) + V_3(z_3)] = \frac{d}{dt}V_1(x_1) + \frac{d}{dt}V_2(z_2) + \frac{d}{dt}V_3(z_3)$$

$$\frac{d}{dt}V_1(x_1) + \frac{d}{dt}V_2(z_2) + \frac{d}{dt}V_3(z_3) = \frac{d}{dt}\left[\frac{1}{2}(x_1)^2\right] + \frac{d}{dt}\left[\frac{1}{2}(z_2)^2\right] + \frac{d}{dt}\left[\frac{1}{2}(z_3)^2\right]$$

$$\dot{V} = x_1\dot{x}_1 + z_1\dot{z}_1 + z_2\dot{z}_2 = x_1[(x_1)^2 - (x_1)^3 + [\phi_1 + z_2]] + z_2[[\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2]$$

$$\dot{V} < 0 \leftrightarrow x_1[(x_1)^2 - (x_1)^3 + [\phi_1 + z_2]] + z_2[[\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[(x_1)^2 - (x_1)^3 + [\phi_1 + z_2]] + z_2[[\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[TERM_1] + z_1[TERM_2] + z_2[TERM_3] < 0$$

If there is any z_i term in first parenthesis, carry them to their corresponding parenthesis by arranging the terms.

Arrange LHS, s.t. there is only x_1 and $\phi_1(x_1)$ terms in TERM-1.

Arrange LHS, s.t. there is not any z_3 terms in TERM-2.

$$x_1[(x_1)^2 - (x_1)^3 + [\phi_1 + z_2]] + z_2[[\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[(x_1)^2 - (x_1)^3 + [\phi_1]] + z_2[x_1 + [\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[(x_1)^2 - (x_1)^3 + [\phi_1]] + z_2[x_1 + [\phi_2] - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

Find $\phi_1(x_1)$ s.t. $[(x_1)^2 - (x_1)^3 + [\phi_1]] = -x_1 - (x_1)^3$

$$[(x_1)^2 - (x_1)^3 + [\phi_1]] = -x_1 - (x_1)^3 \rightarrow \boxed{\phi_1(x_1) = -x_1 - (x_1)^2}$$

Now let us focus on the TERM-2

$$x_1[(x_1)^2 - (x_1)^3 + [\phi_1]] + z_2[x_1 + [\phi_2] - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

$$-[(x_1)^2 + (x_1)^4] - (z_2)^2 - (z_3)^2 < 0$$

Find $\phi_2(x_1, x_2)$ s.t. $[x_1 + [\phi_2] - \dot{\phi}_1] = -z_2$

$$[x_1 + [\phi_2] - \dot{\phi}_1] = -z_2$$

$$\left[x_1 + [\phi_2] - \frac{\partial \phi_1(x_1)}{\partial x_1} \dot{x}_1 \right] = -z_2$$

$$\left[x_1 + [\phi_2] - \frac{\partial [-x_1 - (x_1)^2]}{\partial x_1} \dot{x}_1 \right] = -z_2$$

$$[x_1 + [\phi_2] - [-1 - 2x_1]\dot{x}_1] = -z_2$$

$$\boxed{\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = u \end{array}}$$

$$[x_1 + [\phi_2] - [-1 - 2x_1][(x_1)^2 - (x_1)^3 + x_2]] = -z_2$$

Solve for $\phi_2(x_1, x_2)$

$$\phi_2(x_1, x_2) = -x_1 + [-1 - 2x_1][(x_1)^2 - (x_1)^3 + x_2] - z_2$$

Using $\boxed{z_2 = x_2 - \phi_1(x_1)}$

$$\phi_2(x_1, x_2) = -x_1 + [-1 - 2x_1][(x_1)^2 - (x_1)^3 + x_2] - [x_2 - \phi_1(x_1)]$$

Using $\boxed{\phi_1(x_1) = -x_1 - (x_1)^2}$

$$\phi_2(x_1, x_2) = -x_1 + [-1 - 2x_1][(x_1)^2 - (x_1)^3 + x_2] - [x_2 - [-x_1 - (x_1)^2]]$$

$$\phi_2(x_1, x_2) = -x_1 + [-1 - 2x_1][(x_1)^2 - (x_1)^3 + x_2] - [x_2 + x_1 + (x_1)^2]$$

Now let us focus on the TERM-3

$$x_1[(x_1)^2 - (x_1)^3 + [\phi_1]] + z_2[x_1 + [\phi_2] - \phi_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

$$-[(x_1)^2 + (x_1)^4] - (z_2)^2 - (z_3)^2 < 0$$

Find $u(x_1, x_2, x_3)$ s.t. $[z_2 + u - \dot{\phi}_2] = -z_3$

$$[z_2 + u - \dot{\phi}_2] = -z_3$$

$$\left[z_2 + u - \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} \dot{x}_1 + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} \dot{x}_2 \right] \right] = -z_3$$

$$u = -z_2 + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} \dot{x}_1 + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} \dot{x}_2 \right] - z_3$$

Using $\boxed{z_2 = x_2 - \phi_1(x_1)}$, $\boxed{z_3 = x_3 - \phi_2(x_1, x_2)}$

$$u = -[x_2 - \phi_1(x_1)] + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} \dot{x}_1 + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} \dot{x}_2 \right] - [x_3 - \phi_2(x_1, x_2)]$$

$$\boxed{\begin{array}{l} \dot{x}_1 = (x_1)^2 - (x_1)^3 + x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = u \end{array}}$$

$$u = -[x_2 - \phi_1(x_1)] + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} [(x_1)^2 - (x_1)^3 + x_2] + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} [x_3] \right] - [x_3 - \phi_2(x_1, x_2)]$$

$$u = -[x_2 - \phi_1(x_1)] + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} [(x_1)^2 - (x_1)^3 + x_2] + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} [x_3] \right] - [x_3 - \phi_2(x_1, x_2)]$$

Substitute explicit forms of $\phi_1(x_1)$, $\phi_2(x_1, x_2)$, $\frac{\partial \phi_2(x_1, x_2)}{\partial x_1}$, $\frac{\partial \phi_2(x_1, x_2)}{\partial x_2}$ by using

$$\phi_1(x_1) = -x_1 - (x_1)^2$$

$$\phi_2(x_1, x_2) = -x_1 + [-1 - 2x_1][(x_1)^2 - (x_1)^3 + x_2] - [x_2 + x_1 + (x_1)^2]$$

$$\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} = 2(x_1)^3 - 2x_2 - (2x_1 + 1)(-3(x_1)^2 + 2x_1) - 2(x_1)^2 - 2x_1 - 2$$

$$\frac{\partial \phi_2(x_1, x_2)}{\partial x_2} = -2x_1 - 2$$

[NOTE: This computation is brought to you by MATLAB sym toolbox!]

THE ANSWER:

```
>> [C,T]=coeffs(u,[x1,x2,x3])
```

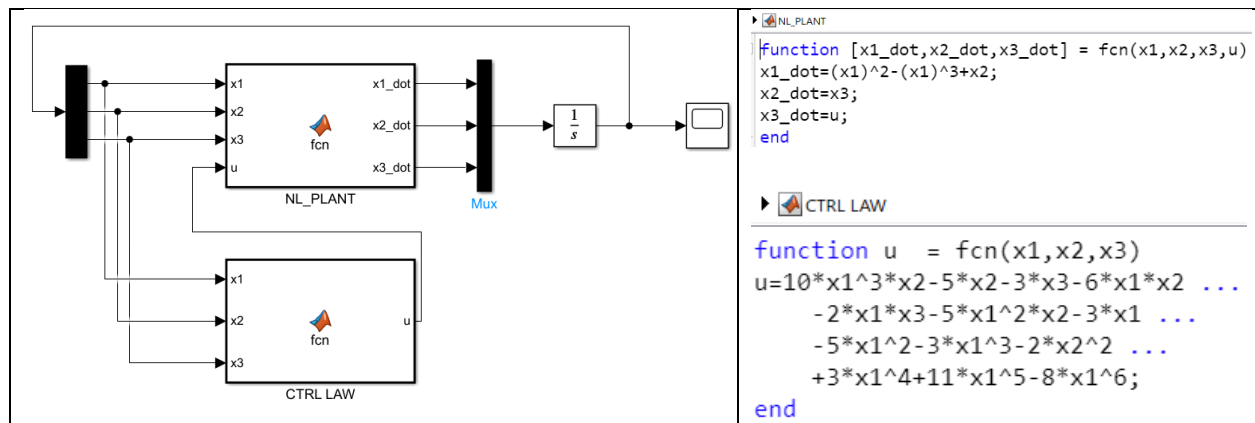
C =

```
[-8, 11, 3, 10, -3, -5, -5, -6, -2, -3, -2, -5, -3]
```

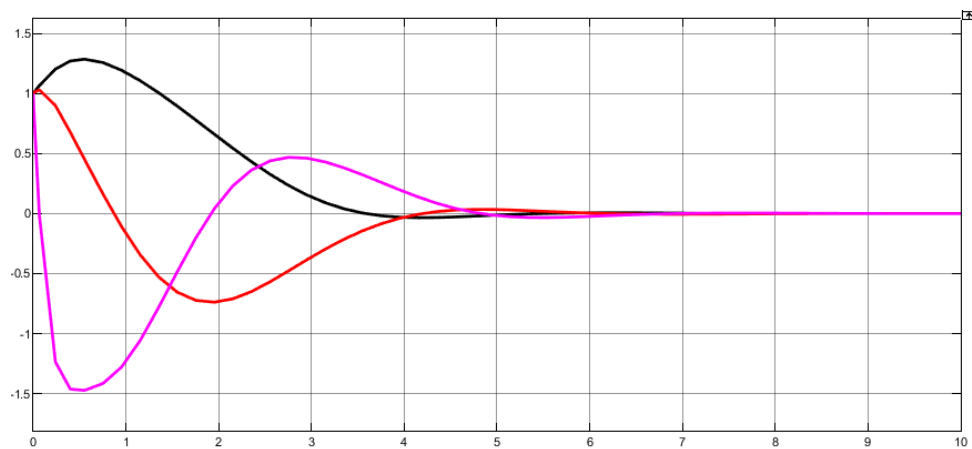
T =

```
[x1^6, x1^5, x1^4, x1^3*x2, x1^3, x1^2*x2, x1^2, x1*x2, x1*x3, x1, x2^2, x2, x3]
```

Check the result by simulating the closed loop system in Simulink[MATLAB].



Result: $[x_1(0), x_2(0), x_3(0)] = [1, 1, 1]$ [initial condition]



Matlab code for problem 1

```

clear all,close all,clc;
%%
syms x1 x2 x3 u real
syms phi1(x1) phi2(x1,x2) real
syms z2 z3 real
x1dot_1=(x1)^2-(x1)^3+x2
x2dot_1=x3
x3dot_1=u
% substitute "phi1+z2" for "x2" in x1dot Eq.
x1dot=subs(x1dot_1,x2,phi1+z2)
% methods(x1dot)
symvar(x1dot) % after subs'n, x1dot should be composed of "[phi1, x1, z2]"
% delete all z2 terms in x1dot and find phi1 s.t. "x1-dyns" is stable
x1dot_wo_z2=subs(x1dot,z2,0)
% at the end the lyap fcn will look like
% V=x1*[x1,phi1,z2]+z2*[phi2,z3,phi1dot]+z3*[u,phi2dot]
% arrange this expr in ...
% V=x1*[x1,phi1]+z2*[phi2,phi1dot]+z3*[u,phi2dot]
% so that
% V=x1*[TERM_1]+z2*[TERM_2]+z3*[TERM_3]
% TERM_1 should not have z2,z3... terms
% TERM_2 should not have z3,z4... terms
z2_terms_to_add_to_z2_paranthesis=(x1dot-x1dot_wo_z2)*x1/z2
symvar(x1dot_wo_z2)
% x1dot=fcn(x1,phi1(x1)), choose "phi1(x1)" s.t. "x1 is stable"
phi1=rhs(isolate(x1dot_wo_z2== -x1-x1^3,phi1))
symvar(phi1) % "phi1" must be composed of [x1]
% find "phi1_dot"
phi1_dot=diff(phi1,x1)*x1dot_1
symvar(phi1_dot) % "phi1_dot" must be composed of [x1,x2]
% x2=z2+phi1 --> x2dot=z2dot+phi1dot --> z2dot=x2dot-phi1dot
% find z2dot expression
z2dot=x2dot_1-phi1_dot
% substitute "phi2+z3" for "x3" in z2dot Eq.
z2dot=subs(z2dot,x3,phi2+z3)
% delete all z3 terms in z2dot and find phi2 s.t. "z2-dyns" is stableXXX
z2dot_wo_z3=subs(z2dot,z3,0)
z3_terms_to_add_to_z3_paranthesis=(z2dot-z2dot_wo_z3)*z2/z3
%
phi2=rhs(isolate(z2dot_wo_z3+z2_terms_to_add_to_z2_paranthesis== -z2,phi2))
symvar(phi2) % "phi2" must be composed of [x1,x2]
phi2=subs(phi2,z2,x2-phi1)
symvar(phi2) % "phi2" must be composed of [x1,x2]
%% isAlways(subs(phi2,z2,x2-phi1)== -x1-(1+2*x1)*(x1^2-x1^3+x2)-(x2+x1+x1^2))
phi2_dot=diff(phi2,x1)*x1dot_1+diff(phi2,x2)*x2dot_1
phi2_dot=simplify(phi2_dot)
symvar(phi2_dot) % "phi2_dot" must be composed of [x1,x2,x3]
% x3=z3+phi2 --> x3dot=z3dot+phi2dot --> z3dot=x3dot-phi2dot
% find z3dot expression
z3dot=x3dot_1-phi2_dot
symvar(z3dot) % "z3dot" must be composed of [x1,x2,x3,u]
u=rhs(isolate(z3dot+z3_terms_to_add_to_z3_paranthesis== -z3,u))
symvar(u) % "u" must be composed of [x1,x2,x3]
u=subs(u,z2,x2-phi1)
u=subs(u,z3,x3-phi2)
symvar(u) % "u" must be composed of [x1,x2,x3]
u=simplify(u)
symvar(u)
[C,T]=coeffs(u,[x1,x2,x3])

```

Matlab code output for problem 1

$$\phi_1(x_1) = -(x_1)^2 - x_1$$

$$\frac{d}{dt}\phi_1(x_1) = -(2x_1 + 1)(-(x_1)^3 + (x_1)^2 + x_2)$$

$$\phi_2(x_1, x_2) = -2x_1 - x_2 - (2x_1 + 1)(-(x_1)^3 + (x_1)^2 + x_2) - (x_1)^2$$

$$\frac{d}{dt}\phi_2(x_1, x_2) = \textit{TOO LONG TO TYPE}$$

```
>> [C,T]=coeffs(phi2_dot,[x1,x2,x3])|
```

```
C =
```

```
[-8, 11, 1, 10, -2, -5, -2, -4, -2, -2, -2, -2]
```

```
T =
```

```
[x1^6, x1^5, x1^4, x1^3*x2, x1^3, x1^2*x2, x1^2, x1*x2, x1*x3, x2^2, x2, x3]
```

$$u(x_1, x_2, x_3) = \textit{TOO LONG TO TYPE}$$

```
>> [C,T]=coeffs(u,[x1,x2,x3])
```

```
C =
```

```
[-8, 11, 3, 10, -3, -5, -5, -6, -2, -3, -2, -5, -3]|
```

```
T =
```

```
[x1^6, x1^5, x1^4, x1^3*x2, x1^3, x1^2*x2, x1^2, x1*x2, x1*x3, x1, x2^2, x2, x3]
```

PROBLEM 2

CONTRIVED EXAMPLE

$$\begin{array}{l} \dot{x}_1 = (x_1)^2 + (x_1 + 1)x_2 \\ \dot{x}_2 = (x_1)^2 + x_3 \\ \dot{x}_3 = u \end{array} \quad \text{Find } u(x_1, x_2, x_3) \text{ s. t. } \boxed{\dot{x} = f_{CL}(x)} \text{ is A.S.}$$

$\begin{array}{l} \dot{x}_1 = (x_1)^2 + (x_1 + 1)x_2 \\ \dot{x}_2 = (x_1)^2 + x_3 \\ \dot{x}_3 = u \end{array}$	$\begin{array}{l} x_2 = \phi_1(x_1) + z_2 \\ \dot{x}_2 = \dot{\phi}_1 + \dot{z}_2 \\ \dot{z}_2 = \dot{x}_2 - \dot{\phi}_1 \end{array}$ Subs $\boxed{\dot{x}_2 = (x_1)^2 + x_3}$ $\dot{z}_2 = [(x_1)^2 + x_3] - \dot{\phi}_1$	$\begin{array}{l} \dot{x}_1 = (x_1)^2 + (x_1 + 1)[\phi_1 + z_2] \\ \dot{z}_2 = [(x_1)^2 + x_3] - \dot{\phi}_1 \\ \dot{x}_3 = u \end{array}$
$\begin{array}{l} \dot{x}_1 = (x_1)^2 + (x_1 + 1)[\phi_1 + z_2] \\ \dot{z}_2 = [(x_1)^2 + x_3] - \dot{\phi}_1 \\ \dot{x}_3 = u \end{array}$	$\begin{array}{l} x_3 = \phi_2(x_1, x_2) + z_3 \\ \dot{x}_3 = \dot{\phi}_2 + \dot{z}_3 \\ \dot{z}_3 = \dot{x}_3 - \dot{\phi}_2 \end{array}$ Subs $\boxed{\dot{x}_3 = u}$ $\dot{z}_3 = u - \dot{\phi}_2$	$\begin{array}{l} \dot{x}_1 = (x_1)^2 + (x_1 + 1)[\phi_1 + z_2] \\ \dot{z}_2 = [(x_1)^2 + [\phi_2 + z_3]] - \dot{\phi}_1 \\ \dot{z}_3 = u - \dot{\phi}_2 \end{array}$

$$\begin{array}{l} \dot{x}_1 = (x_1)^2 + (x_1 + 1)[\phi_1 + z_2] \\ \dot{z}_2 = [(x_1)^2 + [\phi_2 + z_3]] - \dot{\phi}_1 \\ \dot{z}_3 = u - \dot{\phi}_2 \end{array} \rightarrow V(x_1, z_1, z_2) = V_1(x_1) + V_2(z_2) + V_3(z_3)$$

By choosing, $V_1(x_1) = \frac{1}{2}(x_1)^2$, $V_2(z_2) = \frac{1}{2}(z_2)^2$, $V_3(z_3) = \frac{1}{2}(z_3)^2$

[NOTE: more sophisticated functions can also be chosen]

$$\frac{d}{dt}V(x_1, z_1, z_2) = \frac{d}{dt}[V_1(x_1) + V_2(z_2) + V_3(z_3)] = \frac{d}{dt}V_1(x_1) + \frac{d}{dt}V_2(z_2) + \frac{d}{dt}V_3(z_3)$$

$$\frac{d}{dt}V_1(x_1) + \frac{d}{dt}V_2(z_2) + \frac{d}{dt}V_3(z_3) = \frac{d}{dt}\left[\frac{1}{2}(x_1)^2\right] + \frac{d}{dt}\left[\frac{1}{2}(z_2)^2\right] + \frac{d}{dt}\left[\frac{1}{2}(z_3)^2\right]$$

$$\begin{aligned} \dot{V} &= x_1\dot{x}_1 + z_1\dot{z}_1 + z_2\dot{z}_2 \\ &= x_1[(x_1)^2 + (x_1 + 1)[\phi_1 + z_2]] + z_2\left[[x_1)^2 + [\phi_2 + z_3]] - \dot{\phi}_1\right] + z_3[u - \dot{\phi}_2] \end{aligned}$$

$$\dot{V} < 0 \Leftrightarrow x_1[(x_1)^2 + (x_1 + 1)[\phi_1 + z_2]] + z_2\left[[x_1)^2 + [\phi_2 + z_3]] - \dot{\phi}_1\right] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[(x_1)^2 + (x_1 + 1)[\phi_1 + z_2]] + z_2[(x_1)^2 + [\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[TERM_1] + z_1[TERM_2] + z_2[TERM_3] < 0$$

If there is any z_i term in first parenthesis, carry them to their corresponding parenthesis by arranging the terms.

Arrange LHS, s.t. there is only x_1 and $\phi_1(x_1)$ terms in TERM-1.

Arrange LHS, s.t. there is not any z_3 terms in TERM-2.

$$x_1[(x_1)^2 + (x_1 + 1)[\phi_1 + z_2]] + z_2[(x_1)^2 + [\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[(x_1)^2 + (x_1 + 1)[\phi_1]] + z_2[x_1 + (x_1)^2 + [\phi_2 + z_3] - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$$

$$x_1[(x_1)^2 + (x_1 + 1)[\phi_1]] + z_2[x_1 + (x_1)^2 + [\phi_2] - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

Find $\phi_1(x_1)$ s.t. $[(x_1)^2 + (x_1 + 1)[\phi_1]] = -x_1$

$$[(x_1)^2 + (x_1 + 1)[\phi_1]] = -x_1 \rightarrow \boxed{\phi_1(x_1) = -x_1}$$

Now let us focus on the TERM-2

$$x_1[(x_1)^2 + (x_1 + 1)[\phi_1]] + z_2[x_1 + (x_1)^2 + [\phi_2] - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

$$-(x_1)^2 - (z_2)^2 - (z_3)^2 < 0$$

Find $\phi_2(x_1, x_2)$ s.t. $[x_1 + (x_1)^2 + [\phi_2] - \dot{\phi}_1] = -z_2$

$$[x_1 + (x_1)^2 + [\phi_2] - \dot{\phi}_1] = -z_2$$

$$\left[x_1 + (x_1)^2 + [\phi_2] - \frac{\partial \phi_1(x_1)}{\partial x_1} \dot{x}_1 \right] = -z_2$$

$$\left[x_1 + (x_1)^2 + [\phi_2] - \frac{\partial [-x_1]}{\partial x_1} \dot{x}_1 \right] = -z_2$$

$$[x_1 + (x_1)^2 + [\phi_2] - [-1]\dot{x}_1] = -z_2$$

$$[x_1 + (x_1)^2 + [\phi_2] + \dot{x}_1] = -z_2$$

$$\boxed{\begin{aligned} \dot{x}_1 &= (x_1)^2 + (x_1 + 1)x_2 \\ \dot{x}_2 &= (x_1)^2 + x_3 \\ \dot{x}_3 &= u \end{aligned}}$$

$$[x_1 + (x_1)^2 + [\phi_2] + \dot{x}_1] = -z_2$$

$$[x_1 + (x_1)^2 + [\phi_2] + [(x_1)^2 + (x_1 + 1)x_2]] = -z_2$$

Solve for $\phi_2(x_1, x_2)$

$$\phi_2(x_1, x_2) = -x_1 - (x_1)^2 - [(x_1)^2 + (x_1 + 1)x_2] - z_2$$

Using $\boxed{z_2 = x_2 - \phi_1(x_1)}$

$$\phi_2(x_1, x_2) = -x_1 - (x_1)^2 - [(x_1)^2 + (x_1 + 1)x_2] - [x_2 - \phi_1]$$

Using $\boxed{\phi_1(x_1) = -x_1}$

$$\phi_2(x_1, x_2) = -x_1 - (x_1)^2 - [(x_1)^2 + (x_1 + 1)x_2] - [x_2 - [-x_1]]$$

$$\phi_2(x_1, x_2) = -x_1 - (x_1)^2 - [(x_1)^2 + (x_1 + 1)x_2] - [x_2 + x_1]$$

Now let us focus on the TERM-3

$$x_1[(x_1)^2 + (x_1 + 1)[\phi_1]] + z_2[x_1 + (x_1)^2 + [\phi_2] - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

$$-(x_1)^2 - (z_2)^2 - (z_3)^2 < 0$$

Find $u(x_1, x_2, x_3)$ s.t. $[z_2 + u - \dot{\phi}_2] = -z_3$

$$[z_2 + u - \dot{\phi}_2] = -z_3$$

$$\left[z_2 + u - \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} \dot{x}_1 + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} \dot{x}_2 \right] \right] = -z_3$$

$$u = -z_2 + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} \dot{x}_1 + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} \dot{x}_2 \right] - z_3$$

Using $[z_2 = x_2 - \phi_1(x_1)]$, $[z_3 = x_3 - \phi_2(x_1, x_2)]$

$$u = -[x_2 - \phi_1(x_1)] + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} \dot{x}_1 + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} \dot{x}_2 \right] - [x_3 - \phi_2(x_1, x_2)]$$

$$\begin{aligned} \dot{x}_1 &= (x_1)^2 + (x_1 + 1)x_2 \\ \dot{x}_2 &= (x_1)^2 + x_3 \\ \dot{x}_3 &= u \end{aligned}$$

$$u = -[x_2 - \phi_1(x_1)] + \left[\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} [(x_1)^2 + (x_1 + 1)x_2] + \frac{\partial \phi_2(x_1, x_2)}{\partial x_2} [(x_1)^2 + x_3] \right] - [x_3 - \phi_2(x_1, x_2)]$$

Substitute explicit forms of $\phi_1(x_1)$, $\phi_2(x_1, x_2)$, $\frac{\partial \phi_2(x_1, x_2)}{\partial x_1}$, $\frac{\partial \phi_2(x_1, x_2)}{\partial x_2}$ by using

$\phi_1(x_1) = -x_1$	$\phi_2(x_1, x_2) = -x_1 - (x_1)^2 - [(x_1)^2 + (x_1 + 1)x_2] - [x_2 + x_1]$
$\frac{\partial \phi_2(x_1, x_2)}{\partial x_1} = -6x_1 - x_2 - 2$	$\frac{\partial \phi_2(x_1, x_2)}{\partial x_2} = -x_1 - 2$

[NOTE: This computation is brought to you by MATLAB sym toolbox!]

THE ANSWER:

```
>> [C,T]=coeffs(u,[x1,x2,x3])

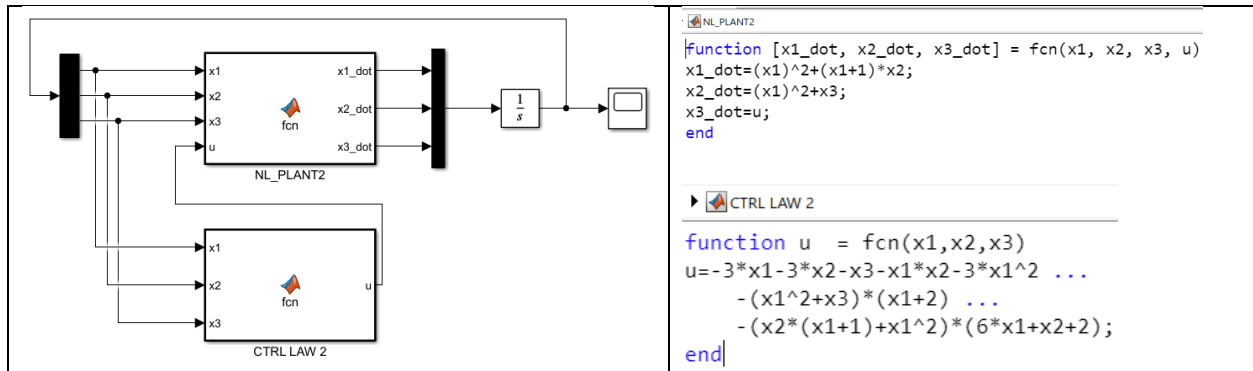
C =

-7, -7, -7, -1, -9, -1, -3, -1, -5, -3]

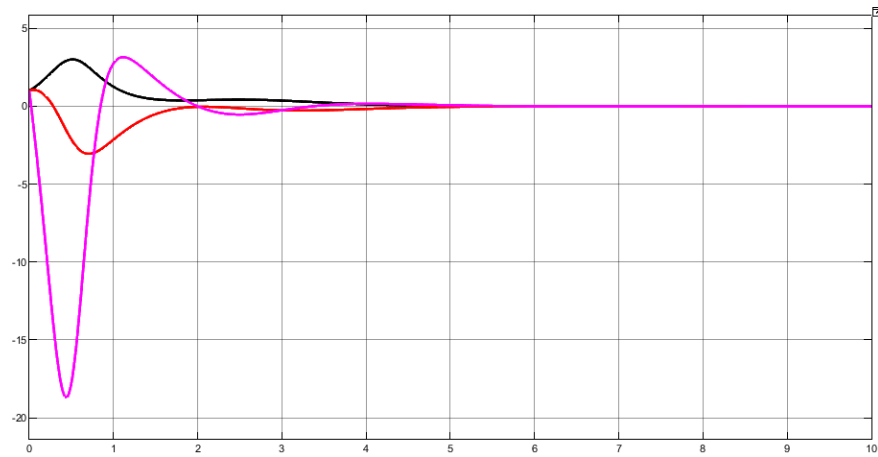
T =

[x1^3, x1^2*x2, x1^2, x1*x2^2, x1*x2, x1*x3, x1, x2^2, x2, x3]
```

Check the result by simulating the closed loop system in Simulink[MATLAB].



Result: $[x_1(0), x_2(0), x_3(0)] = [1, 1, 1]$ [initial condition]



Matlab code for problem 2

```

clear all,close all,clc;
%%
syms x1 x2 x3 u real
syms phi1(x1) phi2(x1,x2) real
syms z2 z3 real
x1dot_1=(x1)^2+(x1+1)*x2
x2dot_1=(x1)^2+x3
x3dot_1=u
% substitute "phi1+z2" for "x2" in x1dot Eq.
x1dot=subs(x1dot_1,x2,phi1+z2)
% methods(x1dot)
symvar(x1dot) % after subs'n, x1dot should be composed of "[phi1, x1, z2]"
% delete all z2 terms in x1dot and find phi1 s.t. "x1-dyns" is stable
x1dot_wo_z2=subs(x1dot,z2,0)
% at the end the lyap fcn will look like
% V=x1*[x1,phi1,z2]+z2*[phi2,z3,phi1dot]+z3*[u,phi2dot]
% arrange this expr in ...
% V=x1*[x1,phi1]+z2*[phi2,phi1dot]+z3*[u,phi2dot]
% so that
% V=x1*[TERM_1]+z2*[TERM_2]+z3*[TERM_3]
% TERM_1 should not have z2,z3... terms
% TERM_2 should not have z3,z4... terms
z2_terms_to_add_to_z2_paranthesis=(x1dot-x1dot_wo_z2)*x1/z2
z2_terms_to_add_to_z2_paranthesis=expand(z2_terms_to_add_to_z2_paranthesis)
symvar(x1dot_wo_z2)
% x1dot=fcn(x1,phi1(x1)), choose "phi1(x1)" s.t. "x1 is stable"
expand(x1dot_wo_z2/x1) % analyze TERM_1
phi1=rhs(isolate(x1dot_wo_z2==-x1,phi1))
% phi1=rhs(isolate(x1dot_wo_z2==-x1-x1^3,phi1))
symvar(phi1) % "phi1" must be composed of [x1]
% find "phi1_dot"
phi1_dot=diff(phi1,x1)*x1dot_1
symvar(phi1_dot) % "phi1_dot" must be composed of [x1,x2]
% x2=z2+phi1 --> x2dot=z2dot+phi1dot --> z2dot=x2dot-phi1dot
% find z2dot expression
z2dot=x2dot_1-phi1_dot
% substitute "phi2+z3" for "x3" in z2dot Eq.
z2dot=subs(z2dot,x3,phi2+z3)
% delete all z3 terms in z2dot and find phi2 s.t. "z2-dyns" is stableXXX
z2dot_wo_z3=subs(z2dot,z3,0)
z3_terms_to_add_to_z3_paranthesis=(z2dot-z2dot_wo_z3)*z2/z3
z3_terms_to_add_to_z3_paranthesis=expand(z3_terms_to_add_to_z3_paranthesis)
%
phi2=rhs(isolate(z2dot_wo_z3+z3_terms_to_add_to_z3_paranthesis==-z2,phi2))
phi2=expand(phi2)
symvar(phi2) % "phi2" must be composed of [x1,x2]
phi2=subs(phi2,z2,x2-phi1)
symvar(phi2) % "phi2" must be composed of [x1,x2]
%% isAlways(subs(phi2,z2,x2-phi1)==-x1-(1+2*x1)*(x1^2-x1^3+x2)-(x2+x1+x1^2))
phi2_dot=diff(phi2,x1)*x1dot_1+diff(phi2,x2)*x2dot_1
pretty(simplify(phi2_dot))
phi2_dot=simplify(phi2_dot)
symvar(phi2_dot) % "phi2_dot" must be composed of [x1,x2,x3]
% x3=z3+phi2 --> x3dot=z3dot+phi2dot --> z3dot=x3dot-phi2dot
% find z3dot expression
z3dot=x3dot_1-phi2_dot
symvar(z3dot) % "z3dot" must be composed of [x1,x2,x3,u]
u=rhs(isolate(z3dot+z3_terms_to_add_to_z3_paranthesis==-z3,u))
symvar(u) % "u" must be composed of [x1,x2,x3]
u=subs(u,z2,x2-phi1)
u=subs(u,z3,x3-phi2)
symvar(u) % "u" must be composed of [x1,x2,x3]
u=simplify(u)
symvar(u)
subs(u,[x1,x2,x3],[0,0,0])
[C,T]=coeffs(u,[x1,x2,x3])

```

Matlab code output for problem 2

$$\phi_1(x_1) = -x_1$$

$$\frac{d}{dt}\phi_1(x_1) = -x_2(x_1 + 1) - (x_1)^2$$

$$\phi_2(x_1, x_2) = -2x_1 - 2x_2 - x_1x_2 - 3(x_1)^2$$

$$\frac{d}{dt}\phi_2(x_1, x_2) = -[(x_1)^2 + x_3](x_1 + 2) - [x_2(x_1 + 1) + (x_1)^2](6x_1 + x_2 + 2)$$

$$u(x_1, x_2, x_3) = \text{TOO LONG TO TYPE}$$

```
>> [C,T]=coeffs(u,[x1,x2,x3])
```

```
C =
```

```
[-7, -7, -7, -1, -9, -1, -3, -1, -5, -3]
```

```
T =
```

```
[x1^3, x1^2*x2, x1^2, x1*x2^2, x1*x2, x1*x3, x1, x2^2, x2, x3]
```


SUMMARY - PROBLEM 3

$\begin{aligned}\dot{x}_1 &= (x_1)^2 + x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= u\end{aligned}$ Variables: (x_1, x_2, x_3)	$\begin{aligned}x_2 &= \phi_1(x_1) + z_2 \\ \dot{x}_2 &= \dot{\phi}_1 + \dot{z}_2 \\ \dot{z}_2 &= \dot{x}_2 - \dot{\phi}_1\end{aligned}$ Subs $\dot{x}_2 = x_3$ $\dot{z}_2 = x_3 - \dot{\phi}_1$	$\begin{aligned}\dot{x}_1 &= (x_1)^2 + (\phi_1 + z_2) \\ \dot{z}_2 &= x_3 - \dot{\phi}_1 \\ \dot{x}_3 &= u\end{aligned}$ Variables: (x_1, z_2, x_3)
$\begin{aligned}\dot{x}_1 &= (x_1)^2 + (\phi_1 + z_2) \\ \dot{z}_2 &= x_3 - \dot{\phi}_1 \\ \dot{x}_3 &= u\end{aligned}$	$\begin{aligned}x_3 &= \phi_2(x_1, x_2) + z_3 \\ \dot{x}_3 &= \dot{\phi}_2 + \dot{z}_3 \\ \dot{z}_3 &= \dot{x}_3 - \dot{\phi}_2\end{aligned}$ Subs $\dot{x}_3 = u$ $\dot{z}_3 = u - \dot{\phi}_2$	$\begin{aligned}\dot{x}_1 &= (x_1)^2 + (\phi_1 + z_2) \\ \dot{z}_2 &= (\phi_2 + z_3) - \dot{\phi}_1 \\ \dot{z}_3 &= u - \dot{\phi}_2\end{aligned}$ Variables: (x_1, z_2, z_3)

Design process $x_1\dot{x}_1 + z_2\dot{z}_2 + z_3\dot{z}_3 < 0$

$x_1[(x_1)^2 + \phi_1 + z_2] + z_2[\phi_2 + z_3 - \dot{\phi}_1] + z_3[u - \dot{\phi}_2] < 0$	
$x_1[(x_1)^2 + \phi_1] + x_1z_2 + z_2[\phi_2 - \dot{\phi}_1] + z_2z_3 + z_3[u - \dot{\phi}_2] < 0$	
$x_1[(x_1)^2 + \phi_1] + z_2[x_1 + \phi_2 - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$	
Sps $W_1(x_1), W_2(z_2), W_3(z_3)$ are pos-def-fcns.	

$$\begin{aligned}[x_1][(x_1)^2 + \phi_1(x_1)] &\leq -W_1(x_1) \\ [z_2][x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &\leq -W_2(z_2) \\ [z_3][z_2 + u - \dot{\phi}_2(x_1, x_2)] &\leq -W_3(z_3)\end{aligned} \rightarrow x_1[(x_1)^2 + \phi_1] + z_2[x_1 + \phi_2 - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

$$\begin{aligned}[x_1][(x_1)^2 + \phi_1(x_1)] &\leq -(x_1)^2 \\ [z_2][x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &\leq -(z_2)^2 \\ [z_3][z_2 + u - \dot{\phi}_2(x_1, x_2)] &\leq -(z_3)^2\end{aligned} \rightarrow x_1[(x_1)^2 + \phi_1] + z_2[x_1 + \phi_2 - \dot{\phi}_1] + z_3[z_2 + u - \dot{\phi}_2] < 0$$

$$\begin{aligned}(x_1)^2 + \phi_1(x_1) &= -x_1 \\ [x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &= -z_2 \\ [z_2 + u - \dot{\phi}_2(x_1, x_2)] &= -z_3\end{aligned} \rightarrow \begin{aligned}[x_1][(x_1)^2 + \phi_1(x_1)] &\leq -(x_1)^2 \\ [z_2][x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &\leq -(z_2)^2 \\ [z_3][z_2 + u - \dot{\phi}_2(x_1, x_2)] &\leq -(z_3)^2\end{aligned}$$

$$\begin{aligned}\phi_1(x_1) &= -x_1 - (x_1)^2 \\ [x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &= -z_2 \\ [z_2 + u - \dot{\phi}_2(x_1, x_2)] &= -z_3\end{aligned} \rightarrow \begin{aligned}[(x_1)^2 + \phi_1(x_1)] &= -x_1 \\ [x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &= -z_2 \\ [z_2 + u - \dot{\phi}_2(x_1, x_2)] &= -z_3\end{aligned}$$

$$\phi_1(x_1) = \frac{\partial \phi_1(x_1)}{\partial x_1} \dot{x}_1 = \frac{\partial [-x_1 - (x_1)^2]}{\partial x_1} [\dot{x}_1] = \frac{\partial [-x_1 - (x_1)^2]}{\partial x_1} [(x_1)^2 + x_2], \quad \begin{aligned}\dot{x}_1 &= (x_1)^2 + x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= u\end{aligned}$$

$$\phi_1(x_1) = [-1 - 2x_1][(x_1)^2 + x_2]$$

$$\begin{aligned}\phi_1(x_1) &= -x_1 - (x_1)^2 \\ [x_1 + \phi_2(x_1, x_2) - \dot{\phi}_1] &= -z_2 \\ [z_2 + u - \dot{\phi}_2(x_1, x_2)] &= -z_3\end{aligned} \rightarrow \begin{aligned}\phi_1(x_1) &= -x_1 - (x_1)^2 \\ [x_1 + \phi_2(x_1, x_2) - [-1 - 2x_1][(x_1)^2 + x_2]] &= -z_2 \\ [z_2 + u - \dot{\phi}_2(x_1, x_2)] &= -z_3\end{aligned}$$

$$\begin{array}{c} \phi_1(x_1) = -x_1 - (x_1)^2 \\ [x_1 + \phi_2(x_1, x_2) - [-1 - 2x_1][(x_1)^2 + x_2]] = -[x_2 - \phi_1] \\ [x_2 - \phi_1] + u - \dot{\phi}_2(x_1, x_2) = -[x_3 - \phi_2] \end{array} \rightarrow \begin{array}{c} \phi_1(x_1) = -x_1 - (x_1)^2 \\ \phi_2 = -x_1 + [-1 - 2x_1][(x_1)^2 + x_2] - [x_2 - \phi_1] \\ [x_2 - \phi_1] + u - \dot{\phi}_2(x_1, x_2) = -[x_3 - \phi_2] \end{array}$$

$$\begin{array}{c} \phi_1(x_1) = -x_1 - (x_1)^2 \\ \phi_2 = -x_1 + [-1 - 2x_1][(x_1)^2 + x_2] - [x_2 - \phi_1] \\ x_2 - \phi_1 + u - \dot{\phi}_2 = -[x_3 - \phi_2] \end{array} \rightarrow \begin{array}{c} \phi_1 = -x_1 - (x_1)^2 \\ \phi_2 = -x_1 + [-1 - 2x_1][(x_1)^2 + x_2] - [x_2 - \phi_1] \\ u = \phi_1 - [x_3 - \phi_2] + \dot{\phi}_2 - x_2 \end{array}$$

Therefore, the ctrl law $u(x_1, x_2, x_3)$ is computed through the algebraic operations. Which is significantly simpler than dealing with PDEs.

PROBLEM 4 Backstepping Control

$\begin{array}{c} \dot{x}_1 = x_2 + \theta_1 x_1 \sin(x_1) \\ \dot{x}_2 = \theta_2 (x_2)^2 + x_1 + u \\ \theta_1 < a, \theta_2 < b \\ \text{Find } u(x_1, x_2) \text{ s.t.} \\ \dot{x} = f_{CL}(x) \text{ is A.S.} \end{array}$	$\begin{array}{c} \dot{x}_1 = x_2 + \theta_1 x_1 \sin(x_1) \\ \dot{x}_2 = \theta_2 (x_2)^2 + x_1 + u \\ \sin(x_1) \leq 1 \\ \theta_1 \sin(x_1) \leq \theta_1 \sin(x_1) < a \end{array}$	$\begin{array}{c} \dot{x}_1 = x_2 + a\delta_1 x_1 \\ \dot{x}_2 = b\delta_2 (x_2)^2 + x_1 + u \\ \delta_1 < 1, \delta_2 < 1 \end{array}$
$\begin{array}{c} \dot{x}_1 = x_2 + a\delta_1 x_1 \\ \dot{x}_2 = b\delta_2 (x_2)^2 + x_1 + u \end{array}$	$\begin{array}{c} x_2 = z_2 + \phi_1(x_1) \\ \dot{x}_2 = \dot{z}_2 + \dot{\phi}_1 \\ \dot{z}_2 = \dot{x}_2 - \dot{\phi}_1 \\ \dot{z}_2 = [b\delta_2 (x_2)^2 + x_1 + u] - \dot{\phi}_1 \end{array}$	$\begin{array}{c} \dot{x}_1 = [z_2 + \phi_1] + a\delta_1 x_1 \\ \dot{z}_2 = b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u \end{array}$

$$x_1 \dot{x}_1 + z_2 \dot{z}_2 < 0$$

$$x_1 [z_2 + \phi_1 + a\delta_1 x_1] + z_2 [b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u] < 0$$

$$x_1 [\phi_1 + a\delta_1 x_1] + z_2 [x_1 + b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u] < 0$$

$$x_1 [\phi_1 + a\delta_1 x_1] + z_2 [2x_1 + b\delta_2 (x_2)^2 - \dot{\phi}_1 + u] < 0$$

$$x_1 [TERM_1] + z_2 [TERM_2] < 0$$

aaa

$$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1), W_1(x_1) \text{ is a pos - def - fcn}$$

$$z_2 [2x_1 + b\delta_2 (x_2)^2 - \dot{\phi}_1 + u] \leq -W_2(z_2), W_2(z_2) \text{ is a pos - def - fcn}$$

aaa

$$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1)$$

$$x_1 \phi_1 + a\delta_1 (x_1)^2 \leq -(x_1)^2$$

$$x_1 \phi_1 \leq -a\delta_1 (x_1)^2 - (x_1)^2$$

$$x_1 \phi_1 \leq -(x_1)^2 [1 + a\delta_1]$$

$$-x_1 \phi_1 \geq (x_1)^2 [1 + a\delta_1]$$

$$-x_1\phi_1 \geq \max_{|\delta_1|<1} (x_1)^2[1+a\delta_1] \geq (x_1)^2[1+a\delta_1]$$

$$-x_1\phi_1 \geq (x_1)^2 \max_{|\delta_1|<1} [1+a\delta_1]$$

$$-x_1\phi_1 \geq (x_1)^2 \left[1 + \max_{|\delta_1|<1} [a\delta_1]\right]$$

$$-x_1\phi_1 \geq (x_1)^2 \left[1 + a \max_{|\delta_1|<1} [\delta_1]\right]$$

$$-x_1\phi_1 \geq (x_1)^2[1+a[1]]$$

$$-x_1\phi_1 \geq (x_1)^2[1+a]$$

$$k := a + 1$$

$$-x_1\phi_1 \geq (x_1)^2k$$

$$-x_1\phi_1 = k(x_1)^2$$

$$-\phi_1 = kx_1$$

$$\phi_1(x_1) = -kx_1$$

aaa

$$z_2[2x_1 + b\delta_2(x_2)^2 - \dot{\phi}_1 + u] \leq -W_2(z_2)$$

$$z_2[2x_1 + b\delta_2(x_2)^2 - \dot{\phi}_1 + u] \leq -(z_2)^2$$

$$z_2 \left[2x_1 + b\delta_2(x_2)^2 - \left[\frac{\partial \phi_1(x_1)}{\partial x_1} [\dot{x}_1] \right] + u \right] \leq -(z_2)^2$$

$$\boxed{\begin{array}{l} \dot{x}_1 = x_2 + a\delta_1x_1 \\ \dot{x}_2 = b\delta_2(x_2)^2 + x_1 + u \end{array}}$$

$$z_2 \left[2x_1 + b\delta_2(x_2)^2 - \left[\frac{\partial [-kx_1]}{\partial x_1} [x_2 + a\delta_1x_1] \right] + u \right] \leq -(z_2)^2$$

$$z_2[2x_1 + b\delta_2(x_2)^2 - [-k[x_2 + a\delta_1x_1]] + u] \leq -(z_2)^2$$

$$z_2[2x_1 + b\delta_2(x_2)^2 + k[x_2 + a\delta_1x_1] + u] \leq -(z_2)^2$$

$$z_2[2x_1 + b\delta_2(x_2)^2 + kx_2 + ka\delta_1x_1 + u] \leq -(z_2)^2$$

$$z_2[2x_1 + kx_2 + u + b\delta_2(x_2)^2 + ka\delta_1x_1] \leq -(z_2)^2$$

$$v := 2x_1 + kx_2 + u$$

$$z_2[v + b\delta_2(x_2)^2 + ka\delta_1x_1] \leq -(z_2)^2$$

$$z_2v + z_2b\delta_2(x_2)^2 + z_2ka\delta_1x_1 \leq -(z_2)^2$$

$$z_2v \leq -(z_2)^2 - z_2b\delta_2(x_2)^2 - z_2ka\delta_1x_1$$

$$\begin{aligned}
-z_2 v &\geq (z_2)^2 + z_2 b \delta_2 (x_2)^2 + z_2 k a \delta_1 x_1 \\
-z_2 v &\geq \max_{|\delta_i| < 1} [(z_2)^2 + z_2 b \delta_2 (x_2)^2 + z_2 k a \delta_1 x_1] \geq (z_2)^2 + z_2 b \delta_2 (x_2)^2 + z_2 k a \delta_1 x_1 \\
-z_2 v &\geq \max_{|\delta_i| < 1} [(z_2)^2] + \max_{|\delta_i| < 1} [z_2 b \delta_2 (x_2)^2] + \max_{|\delta_i| < 1} [z_2 k a \delta_1 x_1] \\
-z_2 v &\geq (z_2)^2 + b(x_2)^2 \max_{|\delta_i| < 1} [z_2 \delta_2] + a k \max_{|\delta_i| < 1} [z_2 \delta_1 x_1] \\
|z_2| \max_{|\delta_i| < 1} [\delta_2] &\geq \max_{|\delta_i| < 1} [z_2 \delta_2] \\
|z_2| |x_1| \max_{|\delta_i| < 1} [\delta_1] &\geq \max_{|\delta_i| < 1} [z_2 \delta_1 x_1] \\
-z_2 v &\geq (z_2)^2 + b(x_2)^2 |z_2| \max_{|\delta_i| < 1} [\delta_2] + a k |z_2| |x_1| \max_{|\delta_i| < 1} [\delta_1] \geq (z_2)^2 + b(x_2)^2 \max_{|\delta_i| < 1} [z_2 \delta_2] + a k \max_{|\delta_i| < 1} [z_2 \delta_1 x_1] \\
-z_2 v &\geq (z_2)^2 + b(x_2)^2 |z_2| \max_{|\delta_i| < 1} [\delta_2] + a k |z_2| |x_1| \max_{|\delta_i| < 1} [\delta_1] \\
-z_2 v &\geq (z_2)^2 + b(x_2)^2 |z_2| [1] + a k |z_2| |x_1| [1] \\
-z_2 v &\geq (z_2)^2 + b(x_2)^2 |z_2| + a k |z_2| |x_1| \\
-z_2 v &\geq (z_2)^2 + b(x_2)^2 |z_2| + a k |z_2| |x_1| \\
z_2 &= |z_2| \operatorname{sgn}(z_2) \\
-|z_2| \operatorname{sgn}(z_2) v &\geq |z_2|^2 + b(x_2)^2 |z_2| + a k |z_2| |x_1| \\
-|z_2| \operatorname{sgn}(z_2) v &\geq |z_2| [|z_2| + b(x_2)^2 + a k |x_1|] \\
-\operatorname{sgn}(z_2) v &\geq [|z_2| + b(x_2)^2 + a k |x_1|] \\
\operatorname{sgn}(z_2) v &\leq -[|z_2| + b(x_2)^2 + a k |x_1|] \\
\operatorname{sgn}(z_2) v &= -[|z_2| + b(x_2)^2 + a k |x_1|] \\
v &= -\left[\frac{1}{\operatorname{sgn}(z_2)}\right] [|z_2| + b(x_2)^2 + a k |x_1|] \\
\boxed{\operatorname{sgn}(z_2) = \frac{1}{\operatorname{sgn}(z_2)}}, \quad \boxed{1 = \frac{1}{1}}, \quad \boxed{-1 = \frac{1}{-1}} \\
v &= -\operatorname{sgn}(z_2) [|z_2| + b(x_2)^2 + a k |x_1|]
\end{aligned}$$

As a result of those operations the control law is computed to be

$$\boxed{
\begin{aligned}
u &= v - (2x_1 + kx_2) \\
k &= a + 1 \\
v &= -\operatorname{sgn}(z_2) [|z_2| + b(x_2)^2 + a k |x_1|] \\
z_2 &= x_2 - \phi_1(x_1) \\
\phi_1(x_1) &= -kx_1
\end{aligned}
}$$

PROBLEM 4 sliding mode control

$\begin{aligned} \dot{x}_1 &= x_2 + \theta_1 x_1 \sin(x_1) \\ \dot{x}_2 &= \theta_2 (x_2)^2 + x_1 + u \\ \theta_1 &< a, \theta_2 < b \\ \text{Find } u(x_1, x_2) \text{ s.t.} \\ \dot{x} &= f_{CL}(x) \text{ is A.S.} \end{aligned}$	$\begin{aligned} \dot{x}_1 &= x_2 + \theta_1 x_1 \sin(x_1) \\ \dot{x}_2 &= \theta_2 (x_2)^2 + x_1 + u \\ \sin(x_1) &\leq 1 \\ \theta_1 \sin(x_1) &\leq \theta_1 \sin(x_1) < a \end{aligned}$	$\begin{aligned} \dot{x}_1 &= x_2 + a\delta_1 x_1 \\ \dot{x}_2 &= b\delta_2 (x_2)^2 + x_1 + u \\ \delta_1 &< 1, \delta_2 < 1 \end{aligned}$
$\begin{aligned} \dot{x}_1 &= x_2 + a\delta_1 x_1 \\ \dot{x}_2 &= b\delta_2 (x_2)^2 + x_1 + u \end{aligned}$	$\begin{aligned} x_2 &= z_2 + \phi_1(x_1) \\ \dot{x}_2 &= \dot{z}_2 + \dot{\phi}_1 \\ \dot{z}_2 &= \dot{x}_2 - \dot{\phi}_1 \\ \dot{z}_2 &= [b\delta_2 (x_2)^2 + x_1 + u] - \dot{\phi}_1 \end{aligned}$	$\begin{aligned} \dot{x}_1 &= [z_2 + \phi_1] + a\delta_1 x_1 \\ \dot{z}_2 &= b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u \end{aligned}$

$$x_1 \dot{x}_1 + z_2 \dot{z}_2 < 0$$

$$x_1 [z_2 + \phi_1 + a\delta_1 x_1] + z_2 [b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u] < 0$$

$$x_1 [\phi_1 + a\delta_1 x_1] + z_2 [x_1 + b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u] < 0$$

$$x_1 [\phi_1 + a\delta_1 x_1] + z_2 [2x_1 + b\delta_2 (x_2)^2 - \dot{\phi}_1 + u] < 0$$

$$x_1 [TERM_1] + z_2 [TERM_2] < 0$$

aaa

$$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1), W_1(x_1) \text{ is a pos-def-fcn}$$

$$z_2 [2x_1 + b\delta_2 (x_2)^2 - \dot{\phi}_1 + u] \leq -W_2(z_2), W_2(z_2) \text{ is a pos-def-fcn}$$

aaa

$$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1)$$

$$x_1 \phi_1 + a\delta_1 (x_1)^2 \leq -(x_1)^2$$

$$x_1 \phi_1 \leq -a\delta_1 (x_1)^2 - (x_1)^2$$

$$x_1 \phi_1 \leq -(x_1)^2 [1 + a\delta_1]$$

$$-x_1 \phi_1 \geq (x_1)^2 [1 + a\delta_1]$$

$$-x_1 \phi_1 \geq \max_{|\delta_1| < 1} (x_1)^2 [1 + a\delta_1] \geq (x_1)^2 [1 + a\delta_1]$$

$$-x_1 \phi_1 \geq (x_1)^2 \max_{|\delta_1| < 1} [1 + a\delta_1]$$

$$-x_1 \phi_1 \geq (x_1)^2 \left[1 + \max_{|\delta_1| < 1} [a\delta_1] \right]$$

$$-x_1 \phi_1 \geq (x_1)^2 \left[1 + a \max_{|\delta_1| < 1} [\delta_1] \right]$$

$$-x_1 \phi_1 \geq (x_1)^2 [1 + a[1]]$$

$$-x_1 \phi_1 \geq (x_1)^2 [1 + a]$$

$$k := a + 1$$

$$-x_1\phi_1 \geq (x_1)^2 k$$

$$-x_1\phi_1 = k(x_1)^2$$

$$-\phi_1 = kx_1$$

$$\phi_1(x_1) = -kx_1$$

By using this, the sliding manifold is computed as

$$x_2 = \phi_1(x_1) \rightarrow x_2 - \phi_1(x_1) = 0 \rightarrow \boxed{s = x_2 - \phi_1(x_1)}$$

Explicitly it can be expressed as

$$s = x_2 - \phi_1(x_1)$$

$$s = x_2 - [-kx_1]$$

$$s = x_2 + kx_1$$

Recall, $\boxed{\dot{s} = -k_1 \text{sgn}(s)}$, $k_1 > 0$ is the general s-dyns that is used to obtain the control law.

Let us look at the s-dyns

$$s = x_2 + kx_1$$

$$\dot{s} = \dot{x}_2 + k\dot{x}_1$$

$$\boxed{\begin{aligned} \dot{x}_1 &= x_2 + a\delta_1 x_1 \\ \dot{x}_2 &= b\delta_2(x_2)^2 + x_1 + u \end{aligned}}$$

$$\dot{s} = [b\delta_2(x_2)^2 + x_1 + u] + k[x_2 + a\delta_1 x_1]$$

$$\dot{s} = b\delta_2(x_2)^2 + x_1 + u + kx_2 + ak\delta_1 x_1$$

Let us gather the “known terms” together and rename them,

$$\dot{s} = b\delta_2(x_2)^2 + \boxed{x_1 + u + kx_2} + ak\delta_1 x_1$$

$$v := x_1 + u + kx_2$$

$$\dot{s} = b\delta_2(x_2)^2 + v + ak\delta_1 x_1$$

Now we have $\boxed{\dot{s} = [\text{unc - terms}] + v}$ form that is easy to manipulate.

Let us gather the “unknown terms” together and rename them,

$$\dot{s} = b\delta_2(x_2)^2 + v + ak\delta_1 x_1$$

$$\dot{s} = b\delta_2(x_2)^2 + ak\delta_1 x_1 + v$$

$$\dot{s} = \boxed{b\delta_2(x_2)^2 + ak\delta_1 x_1} + v$$

$$\Delta(x_1, x_2) := b\delta_2(x_2)^2 + ak\delta_1x_1$$

$$\dot{s} = \Delta(x_1, x_2) + v$$

Recall, the objective is to have a resulting s-dyns of $\boxed{\dot{s} = -k_1 \text{sgn}(s)}, k_1 > 0$

The uncertain term $\Delta(x_1, x_2)$ is also known to be bounded since $\boxed{|\delta_i| < 1}$, this info can be used to determine the upper bound for $\Delta(x_1, x_2)$, this can be represented as

$$|\Delta(x_1, x_2)| \leq \beta(x)$$

As a result of this, the s-dyns can be represented as

$$\boxed{\dot{s} = \Delta(x_1, x_2) + v}, \boxed{|\Delta(x_1, x_2)| \leq \beta(x)}$$

To stabilize this dyns, the proposed control law is $\boxed{v = -\beta(x)\text{sgn}(s)}$

Question

$$\boxed{\dot{s} = \Delta(x_1, x_2) + v}, \boxed{|\Delta(x_1, x_2)| \leq \beta(x)}$$

Find v such that s-dyns are stable [finite time stability].

Answer: $\boxed{v = -\beta(x)\text{sgn}(s)}$

$$\dot{s} = \Delta(x) - \beta(x)\text{sgn}(s)$$

Let us define

$$|\Delta(x)| < \Delta_{max}$$

Therefore, $\boxed{\Delta(x) = \Delta_{max}\delta}, |\delta| < 1$

$$\dot{s} = \Delta(x) - \beta(x)\text{sgn}(s)$$

$$\dot{s} = \Delta_{max}\delta - \beta(x)\text{sgn}(s)$$

The uncertainty term can be expressed as $\boxed{\delta = \delta_2\text{sgn}(s)}$

[idk how to prove than other than appealing to intuition, but there must be “uncertain term algebra” type of course to make these operations more systematic, [maybe there is and idk, but if there is no such a course, there should be]]

$$\dot{s} = \Delta_{max}\delta - \beta(x)\text{sgn}(s)$$

$$\dot{s} = \Delta_{max}\delta_2\text{sgn}(s) - \beta(x)\text{sgn}(s)$$

$$\dot{s} = [\Delta_{max}\delta_2 - \beta(x)]\text{sgn}(s)$$

$$\dot{s} = [TERM]\text{sgn}(s)$$

The TERM should be negative, therefore

$$\Delta_{max}\delta_2 - \beta(x) < 0$$

$$\Delta_{max}\delta_2 < \beta(x)$$

$$\Delta_{max}\delta_2 < \Delta_{max} \max_{|\delta_2| < 1} [\delta_2] < \beta(x)$$

$$\Delta_{max}\delta_2 < \Delta_{max}[1] < \beta(x)$$

$$\Delta_{max}\delta_2 < \Delta_{max} < \beta(x)$$

$$\Delta_{max} < \beta(x)$$

Therefore, the suggested control law has the ability to drive the states onto the sliding manifold in finite time.

Let us determine $\Delta_{max}(x_1, x_2)$ term explicitly,

$$\Delta(x) = b\delta_2(x_2)^2 + ak\delta_1x_1$$

$$\Delta(x) \leq \Delta_{max}(x)$$

$$b\delta_2(x_2)^2 + ak\delta_1x_1 \leq \Delta_{max}(x)$$

$$b\delta_2(x_2)^2 + ak\delta_1x_1 \leq \max_{|\delta_2| < 1} [b\delta_2(x_2)^2] + \max_{|\delta_1| < 1} [ak\delta_1x_1] \leq \Delta_{max}(x)$$

$$b\delta_2(x_2)^2 + ak\delta_1x_1 \leq b(x_2)^2 \max_{|\delta_2| < 1} [\delta_2] + ak \max_{|\delta_1| < 1} [\delta_1x_1] \leq \Delta_{max}(x)$$

$$|x_1| \max_{|\delta_1| < 1} [\delta_1] \geq \max_{|\delta_1| < 1} [\delta_1x_1]$$

$$b(x_2)^2 \max_{|\delta_2| < 1} [\delta_2] + ak \max_{|\delta_1| < 1} [\delta_1x_1] \leq b(x_2)^2 \max_{|\delta_2| < 1} [\delta_2] + ak|x_1| \max_{|\delta_1| < 1} [\delta_1] \leq \Delta_{max}(x)$$

$$b(x_2)^2 \max_{|\delta_2| < 1} [\delta_2] + ak|x_1| \max_{|\delta_1| < 1} [\delta_1] \leq \Delta_{max}(x)$$

$$b(x_2)^2[1] + ak|x_1|[1] \leq \Delta_{max}(x)$$

$$b(x_2)^2 + ak|x_1| \leq \Delta_{max}(x)$$

Let us determine the condition on $\beta(x)$

$$\Delta_{max}(x) < \beta(x)$$

$$\Delta_{max}(x) + \beta_0 = \beta(x), \boxed{\beta_0 > 0}$$

Finally, the control law is written as

$$\boxed{u = -x_1 - kx_2 + v} \rightarrow \boxed{v = -\beta(x)sgn(s)} \rightarrow \boxed{\beta(x) = \Delta_{max}(x) + \beta_0} \rightarrow \boxed{\Delta_{max}(x) = b(x_2)^2 + ak|x_1|}$$

$$\boxed{k > 0}$$

$$\boxed{\beta_0 > 0}$$

Ctrl law can be represented as

$$\boxed{\begin{array}{l} k > 0 \\ \beta_0 > 0 \\ \Delta_{max}(x) = b(x_2)^2 + ak|x_1| \\ s = x_2 + kx_1 \\ \beta(x) = \Delta_{max}(x) + \beta_0 \\ v = -\beta(x)sgn(s) \\ u = -x_1 - kx_2 + v \end{array}}$$

PROBLEM 4 systematizing the problem

$\dot{x}_1 = x_2 + \theta_1 x_1 \sin(x_1)$ $\dot{x}_2 = \theta_2 (x_2)^2 + x_1 + u$ $ \theta_1 < a, \theta_2 < b$ Find $u(x_1, x_2)$ s.t. $\dot{x} = f_{CL}(x)$ is A.S.	$\dot{x}_1 = x_2 + \theta_1 x_1 \sin(x_1)$ $\dot{x}_2 = \theta_2 (x_2)^2 + x_1 + u$ $ \sin(x_1) \leq 1$ $ \theta_1 \sin(x_1) \leq \theta_1 \sin(x_1) < a$	$\dot{x}_1 = x_2 + a\delta_1 x_1$ $\dot{x}_2 = b\delta_2 (x_2)^2 + x_1 + u$ $ \delta_1 < 1, \delta_2 < 1$
$\dot{x}_1 = x_2 + a\delta_1 x_1$ $\dot{x}_2 = b\delta_2 (x_2)^2 + x_1 + u$	$x_2 = z_2 + \phi_1(x_1)$ $\dot{x}_2 = \dot{z}_2 + \dot{\phi}_1$ $\dot{z}_2 = \dot{x}_2 - \dot{\phi}_1$ $\dot{z}_2 = [b\delta_2 (x_2)^2 + x_1 + u] - \dot{\phi}_1$	$\dot{x}_1 = [z_2 + \phi_1] + a\delta_1 x_1$ $\dot{z}_2 = b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u$ $\dot{x}_1 = [z_2 + \phi_1] + a\delta_1 x_1$ $\dot{z}_2 = b\delta_2 (z_2 + \phi_1)^2 + x_1 - \dot{\phi}_1 + u$

$$x_1 \dot{x}_1 + z_2 \dot{z}_2 < 0$$

$$x_1 [z_2 + \phi_1 + a\delta_1 x_1] + z_2 [b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u] < 0$$

$$x_1 [\phi_1 + a\delta_1 x_1] + z_2 [x_1 + b\delta_2 (x_2)^2 + x_1 - \dot{\phi}_1 + u] < 0$$

$$x_1 [\phi_1 + a\delta_1 x_1] + z_2 [2x_1 + b\delta_2 (x_2)^2 - \dot{\phi}_1 + u] < 0$$

$$x_1 [TERM_1] + z_2 [TERM_2] < 0$$

aaa

$$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1), W_1(x_1) \text{ is a pos-def-fcn}$$

$$z_2 [2x_1 + b\delta_2 (x_2)^2 - \dot{\phi}_1 + u] \leq -W_2(z_2), W_2(z_2) \text{ is a pos-def-fcn}$$

aaa

$$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1)$$

$x_1 [\phi_1 + a\delta_1 x_1] \leq -W_1(x_1)$	Choose $W_1(x_1) = -(x_1)^2$
$x_1 [\phi_1 + a\delta_1 x_1] \leq -(x_1)^2$	Distribute x_1 over the parenthesis
$x_1 \phi_1 + a\delta_1 (x_1)^2 \leq -(x_1)^2$	Isolate the term with ϕ_1
$x_1 \phi_1 \leq -a\delta_1 (x_1)^2 - (x_1)^2$	Arrange the terms st $f(\phi_1) \geq \text{other terms}$
$-x_1 \phi_1 \geq a\delta_1 (x_1)^2 + (x_1)^2$	Find a known upper bound for LHS
$-x_1 \phi_1 \geq \text{upper}\{a\delta_1 (x_1)^2 + (x_1)^2\} \geq a\delta_1 (x_1)^2 + (x_1)^2$	“upper” is a made-up operator: $\text{upper}\{k(x)f(\delta_{unc})\} = k(x) \max_{ \delta_i < 1} \{f(\delta_{unc})\}$
$-x_1 \phi_1 \geq \text{upper}\{a\delta_1 (x_1)^2 + (x_1)^2\}$ $-x_1 \phi_1 \geq \text{upper}\{a\delta_1 (x_1)^2\} + \text{upper}\{(x_1)^2\}$ $-x_1 \phi_1 \geq a(x_1)^2 \max_{ \delta_i < 1} \{\delta_1\} + (x_1)^2$ $-x_1 \phi_1 \geq a(x_1)^2 [1] + (x_1)^2$ $-x_1 \phi_1 \geq a(x_1)^2 + (x_1)^2$	If there is no unc-term that is left in the LHS, turn the ineq to eq
$-x_1 \phi_1 \geq a(x_1)^2 + (x_1)^2$ $-x_1 \phi_1 = a(x_1)^2 + (x_1)^2$	Solve for $\phi_1(x_1)$
$-x_1 \phi_1 = a(x_1)^2 + (x_1)^2$	Define $k := a + 1$ for simplicity

$\begin{aligned} -\phi_1 &= ax_1 + x_1 \\ \phi_1 &= -ax_1 - x_1 \\ \phi_1 &= -x_1(a+1) \end{aligned}$	
$\phi_1 = -kx_1$	

Focus on time-der-phi1

$\frac{d}{dt}\phi_1(x_1) = \frac{\partial\phi_1(x_1)}{\partial x_1}\dot{x}_1$	$\begin{aligned} \dot{x}_1 &= x_2 + a\delta_1x_1 \\ \dot{x}_2 &= b\delta_2(x_2)^2 + x_1 + u \end{aligned}$
$\dot{\phi}_1 = \frac{\partial\phi_1(x_1)}{\partial x_1}\dot{x}_1$	$\dot{\phi}_1 = \frac{\partial[-kx_1]}{\partial x_1}[x_2 + a\delta_1x_1]$
$\phi_1 = -k[x_2 + a\delta_1x_1]$	$\dot{\phi}_1 = -kx_2 - ka\delta_1x_1$

Focus on u-term

$z_2[2x_1 + b\delta_2(x_2)^2 - \dot{\phi}_1 + u] \leq -W_2(z_2)$	Choose $W_2(z_2) = (z_2)^2$
$z_2[2x_1 + b\delta_2(x_2)^2 - \dot{\phi}_1 + u] \leq -(z_2)^2$	Subs explicit form of $\dot{\phi}_1$
$\begin{aligned} z_2[2x_1 + b\delta_2(x_2)^2 - [-kx_2 - ka\delta_1x_1] + u] &\leq -(z_2)^2 \\ z_2[2x_1 + b\delta_2(x_2)^2 + kx_2 + ka\delta_1x_1 + u] &\leq -(z_2)^2 \end{aligned}$	Gather the known-terms in parenthesis and rename them
$\begin{aligned} z_2[2x_1 + kx_2 + u + b\delta_2(x_2)^2 + ka\delta_1x_1] &\leq -(z_2)^2 \\ z_2[\boxed{2x_1 + kx_2 + u} + \boxed{b\delta_2(x_2)^2 + ka\delta_1x_1}] &\leq -(z_2)^2 \\ z_2[\boxed{v} + \boxed{b\delta_2(x_2)^2 + ka\delta_1x_1}] &\leq -(z_2)^2 \end{aligned}$	distribute z_2 over the parenthesis
$\boxed{z_2v + z_2[b\delta_2(x_2)^2 + ka\delta_1x_1]} \leq -(z_2)^2$	Isolate v-term
$z_2v \leq -(z_2)^2 - z_2[b\delta_2(x_2)^2 + ka\delta_1x_1]$	Put it in the v-term $\geq$ other terms
$-z_2v \geq (z_2)^2 + z_2[b\delta_2(x_2)^2 + ka\delta_1x_1]$	Gather the unc-terms $f(u) \geq f_{unc}(\delta) + f_{known}(x, z)$
$-z_2v \geq (z_2)^2 + z_2b\delta_2(x_2)^2 + z_2ka\delta_1x_1$	Use "upper-operator" for LHS
$\begin{aligned} -z_2v &\geq upper\{(z_2)^2 + z_2b\delta_2(x_2)^2 + z_2ka\delta_1x_1\} \\ -z_2v &\geq upper\{(z_2)^2\} + upper\{z_2b\delta_2(x_2)^2 + z_2ka\delta_1x_1\} \\ -z_2v &\geq upper\{(z_2)^2\} + upper\{z_2b\delta_2(x_2)^2\} + upper\{z_2ka\delta_1x_1\} \end{aligned}$	Fcn of upper-operator: $\begin{aligned} &upper\{f_{known}(x, z)f_{unc}(\delta)\} \\ &= f_{known}(x, z) \max_{ \delta_i < 1} \{f(\delta_{unc})\} \end{aligned}$
$\begin{aligned} -z_2v &\geq (z_2)^2 + b(x_2)^2 z_2 \max_{ \delta_i < 1} \{\delta_2\} + ak x_1z_2 \max_{ \delta_i < 1} \{\delta_1\} \\ -z_2v &\geq (z_2)^2 + b(x_2)^2 z_2 [1] + ak x_1z_2 [1] \\ -z_2v &\geq (z_2)^2 + b(x_2)^2 z_2 + ak x_1 z_2 \end{aligned}$	
$\begin{aligned} - z_2 sgn(z_2)v &\geq z_2 ^2 + b(x_2)^2 z_2 + ak x_1 z_2 \\ -sgn(z_2)v &\geq z_2 + b(x_2)^2 + ak x_1 \end{aligned}$	Use $\boxed{z = z sgn(z)}$ to isolate v
$\begin{aligned} -v &\geq \left[\frac{1}{sgn(z_2)}\right][z_2 + b(x_2)^2 + ak x_1] \\ -v &\geq [sgn(z_2)][z_2 + b(x_2)^2 + ak x_1] \\ v &\leq -[sgn(z_2)][z_2 + b(x_2)^2 + ak x_1] \end{aligned}$	Use $\boxed{\frac{1}{sgn(z)} = sgn(z)}$ to isolate v
$v = -[sgn(z_2)][z_2 + b(x_2)^2 + ak x_1]$	Turn the ineq to eq to find v

Finally,

$$2x_1 + kx_2 + u = v \rightarrow u = v - [2x_1 + kx_2]$$

$$v = -[sgn(z_2)][|z_2| + b(x_2)^2 + ak|x_1|]$$

$$u = [-\text{sgn}(z_2)][|z_2| + b(x_2)^2 + ak|x_1|] - [2x_1 + kx_2]$$

By using $\boxed{z_2 = x_2 - \phi_1(x_1)}$

$$u = [-\text{sgn}([x_2 - \phi_1(x_1)])][|x_2 - \phi_1(x_1)| + b(x_2)^2 + ak|x_1|] - [2x_1 + kx_2]$$

By using the explicit form of ϕ_1

$$u = [-\text{sgn}([x_2 - \phi_1(x_1)])][|x_2 - \phi_1(x_1)| + b(x_2)^2 + ak|x_1|] - [2x_1 + kx_2]$$

$$\phi_1 = -kx_1$$

$$\boxed{u = [-\text{sgn}([x_2 + kx_1])][|x_2 + kx_1| + b(x_2)^2 + ak|x_1|] - [2x_1 + kx_2]}$$

This control law can be checked by simulating the following dynamical system,

$$\begin{cases} \dot{x}_1 = x_2 + a\delta_1 x_1 \\ \dot{x}_2 = b\delta_2(x_2)^2 + x_1 + u \end{cases}$$

$$|\delta_1| < 1, |\delta_2| < 1$$

$$\boxed{u = [-\text{sgn}([x_2 + kx_1])][|x_2 + kx_1| + b(x_2)^2 + ak|x_1|] - [2x_1 + kx_2]}, \boxed{k = a + 1}$$

$$\begin{cases} \dot{x}_1 = x_2 + a\delta_1 x_1 \\ \dot{x}_2 = b\delta_2(x_2)^2 + x_1 + u \end{cases}$$

Matlab code [to simulate the CL-system]

```

clear all,close all,clc;
%%
a=1;b=2;delta_1=0.1;delta_2=-0.2;
%% single run
% % % tspan = [0 5]; x0 = [1 1];
% % % [t,x] = ode45(@(t,x) odefcn(t,x,a,b,delta_1,delta_2), tspan, x0);
% % % plot(x)
%% multiple runs
tspan = [0 5];
x0 = [1 2];
fig1=figure(1);fig1.Color=[1,1,1];
ax1=axes('Parent',fig1);
fig2=figure(2);fig2.Color=[1,1,1];
ax2=axes('Parent',fig2);
%%
for ii=1:1:10
%   plot(randn(100,1))
    sigma=0.2; mu=0;
    delta_1=mu+randn(1)*sigma; if(abs(delta_1)>1) delta_1=0.97; end
    delta_2=mu+randn(1)*sigma; if(abs(delta_2)>1) delta_2=0.97; end
    [t,x] = ode45(@(t,x) odefcn(t,x,a,b,delta_1,delta_2), tspan, x0);
    x1_vec=x(:,1); x2_vec=x(:,2);
    set(0, 'currentfigure', fig1);
    set(fig1, 'currentaxes', ax1);
    plot(t,x1_vec,"Color",[1,0,0],"LineStyle","--","LineWidth",[1],"Parent",ax1); hold on;
    plot(t,x2_vec,"Color",[0,0,1],"LineStyle","-","LineWidth",[1],"Parent",ax1); hold on;
    set(0, 'currentfigure', fig2);
    set(fig2, 'currentaxes', ax2);
    plot(x1_vec,x2_vec,"Color",[0,0,1],"LineStyle","-","LineWidth",[1],"Parent",ax2); hold on;
end
% ax1 properties
ax1.TickLabelInterpreter='latex';
ax1.XLabel.Interpreter='latex';
ax1.YLabel.Interpreter='latex';
ax1.XLabel.String='time';
ax1.YLabel.String='x(t)';
ax1.YLabel.FontSize=[20];
ax1.FontSize=[20];
ax1.XGrid='on';
ax1.YGrid='on';
% ax2 properties
ax2.TickLabelInterpreter='latex';
ax2.XLabel.Interpreter='latex';
ax2.YLabel.Interpreter='latex';
ax2.XLabel.String='$x_{1}(t)$';
ax2.YLabel.String='$x_{2}(t)$';
ax2.YLabel.FontSize=[20];
ax2.FontSize=[20];
ax2.XGrid='on';
ax2.YGrid='on';
legend(ax1,{'$x_{1}(t)$','$x_{2}(t)$'},"Interpreter","latex");
%%
function xdot = odefcn(t,x,a,b,delta_1,delta_2)
    xdot = zeros(2,1);
    x_1=x(1); x_2=x(2);
    k=a+1;
    u=[-sign([x_2+k*x_1])*[abs([x_2+k*x_1])+b*(x_2)^2+a*k*abs(x_1)]]-[2*x_1+k*x_2];
    xdot(1) = x_2+a*delta_1*x_1;
    xdot(2) = b*delta_2*(x_2)^2+x_1+u;
end

```

Matlab code output[CL-system simulation results]

